Staphylococcus saprophyticus and Enterococcus

GUS-3rd year

LG2

Dr. Samira Fattah

PhD Medical Bacteriology

samira.hamid@hmu.edu.krd

Learning objectives

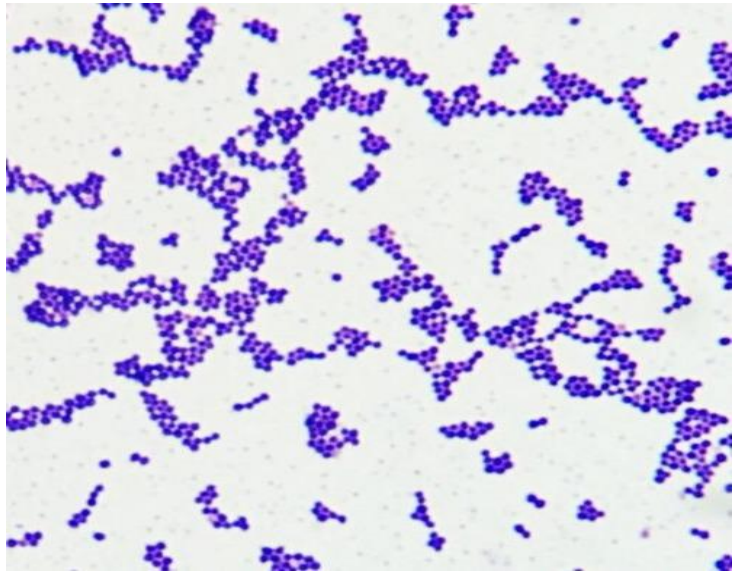
- Recognize the main characteristics of both organisms.
- Explain colonization, virulence, and the clinical syndromes.
- Interpret urine cultures and key identification tests.
- Choose treatment principles using the infection site and susceptibility results.
- Explain VRE prevention and when bacteriuria needs treatment.

Clinical importance of these Gram-positive cocci

Feature	***S. saprophyticus***	***Enterococcus* species**
Key character	Coagulase-negative staphylococcus	Normal intestinal colonizer
Occurrence	Classically in young sexually active women	Important opportunistic hospital pathogen
Important disease	Major cause of uncomplicated cystitis	UTI, bloodstream infection, endocarditis

Staphylococcus saprophyticus
Main characteristics

- Gram-positive cocci, usually in **clusters**.
- **Catalase** positive
- C**oagulase** negative.
- Facultative anaerobe, nonmotile, and non-spore-forming.
- Usually **nonhemolytic** on blood agar.
- Naturally resistant to **novobiocin**, useful presumptive feature.



Staphylococcal clusters on Gram stain

Reservoir and epidemiology

- Colonizes the perineal and genitourinary region and may occur in the gastrointestinal tract.
- An important cause of **community-acquired cystitis** in young, sexually active women.
- Often described as the **second** commonest **cause** of UTI **after *E. coli***.
- Infection usually follows ascent from periurethral flora.

Virulence factors:

- Surface **adhesins** support attachment to uroepithelial cells.
- **Biofilm** formation supports persistence on surfaces in some infections.

Clinical presentation
Acute cystitis

- Dysuria (burning pain during urination).
- Urinary urgency and increased frequency.
- Suprapubic discomfort; sometimes hematuria.
- Uncomplicated cystitis: usually no fever or flank pain.

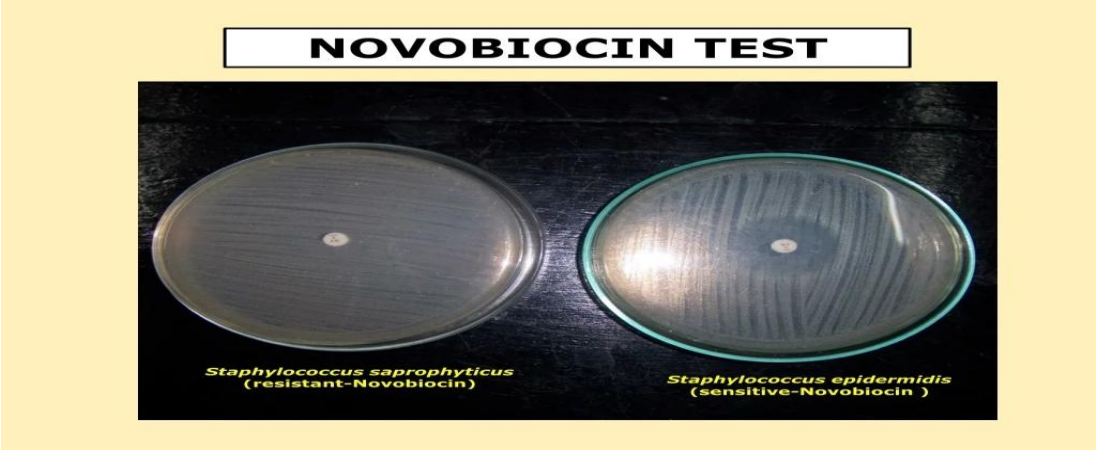
Fever, rigors or flank pain suggest upper/systemic UTI.

Diagnosis of S. saprophyticus cystitis

- Clinical findings: dysuria, urgency, frequency and absence of systemic signs.
- Urinalysis: pyuria / **leukocyte esterase** often present.
- Urine **nitrite** may be **negative** (does not typically reduce nitrate).
- Urine culture when indicated; interpret colony counts with symptoms.
- Confirm identification and AST by validated laboratory methods.

Lab Identification of S. saprophyticus

Test	Typical finding	Interpretation
Gram stain	Positive cocci in clusters	Suggests staphylococci
Catalase	Positive	Supports genus identification
Tube coagulase	Negative	Coagulase-negative staphylococcus
Novobiocin	Resistant	Supports *S. saprophyticus*
Vitek 2 system or validated system	Species identification	Confirms clinically significant isolates

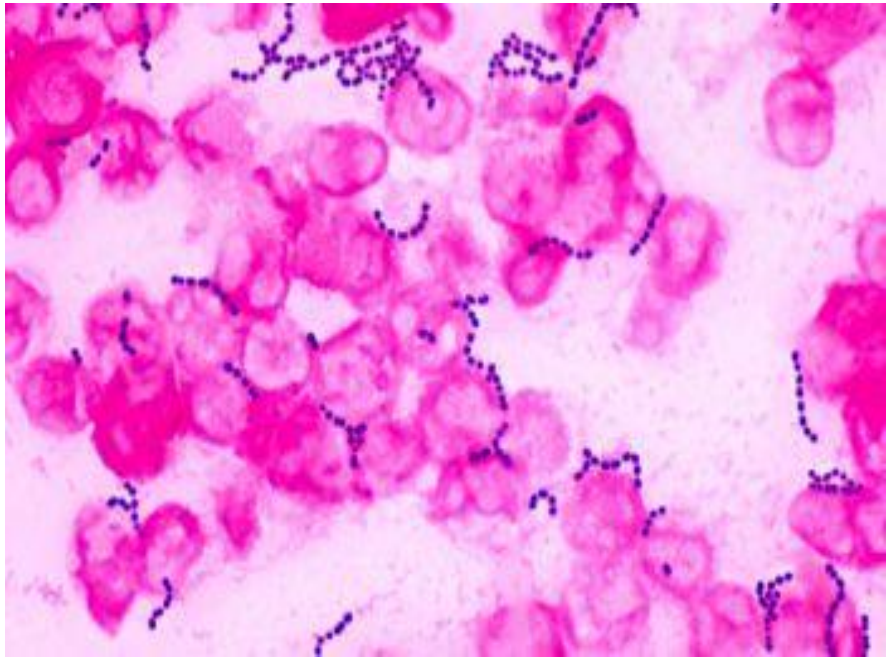


Treatment of S. saprophyticus cystitis

- For localized uncomplicated cystitis: **nitrofurantoin** is a common first-line option.
- Adjust for pregnancy, kidney function, allergy and complicated disease.
- Nitrofurantoin must not be used for pyelonephritis or bacteremia.

Enterococcus
Main characteristics

- Gram-positive cocci, usually in pairs or short chains.
- Facultative anaerobes; catalase **negative** in routine identification.
- Part of normal human **intestinal** flora.
- Historically classified with group D streptococci.



Species of Medical Importance

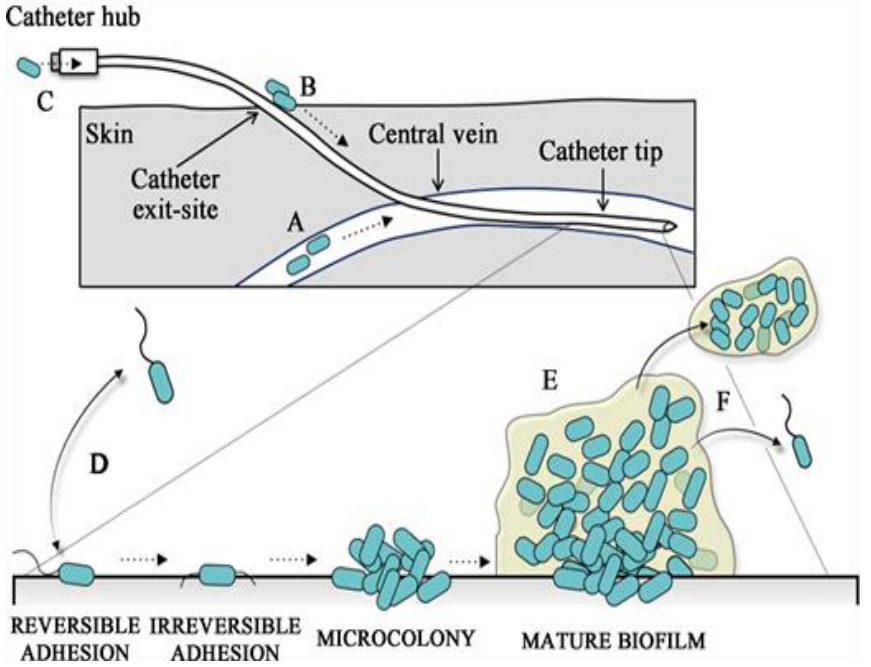
- *Enterococcus faecalis*
- *Enterococcus faecium*

Reservoir and transmission

- The **gastrointestinal** tract is the main **reservoir**.
- Many infections arise from the patient's own flora.
- Healthcare spread occurs through contaminated hands, equipment, and surfaces.
- Risk increases with urinary **catheters**, **prolonged** admission, **antibiotic** exposure, or **impaired** host defenses.

Enterococcal virulence and persistence

- **Adhesins** facilitate attachment to host tissue and devices.
- **Biofilm** supports persistence on catheters and prosthetic material.
- Some strains produce **cytolysin** and **gelatinase**.
- Environmental tolerance helps survival in healthcare settings.
- Resistance to **multiple antibiotics** complicates treatment.



Major clinical infections caused by Enterococcus

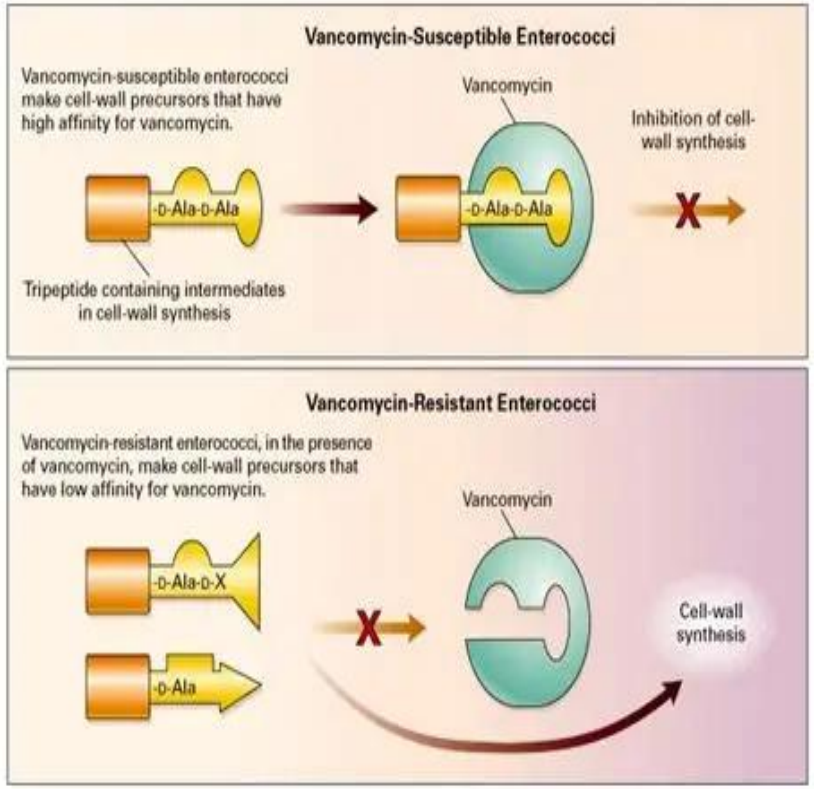
- Urinary tract infection, especially with urinary devices.
- Bloodstream infection, including catheter-related bacteremia.
- Infective endocarditis, especially *E. faecalis*.
- Intra-abdominal, pelvic and postoperative wound infection.
- Often healthcare-associated, particularly with *E. faecium*.

Why enterococci are difficult to treat

- **Intrinsic resistance** makes cephalosporin monotherapy ineffective.
- **Low-level aminoglycoside resistance** limits aminoglycoside monotherapy.
- Acquired **high-level aminoglycoside resistance** can abolish synergistic killing.
- Altered penicillin-binding proteins can cause **ampicillin resistance**.
- The infection site and AST result determine the treatment approach.

Acquired resistance and VRE

- **Vancomycin-resistant Enterococcus** (VRE) is a major hospital threat.
- *E. faecium* has a particularly high burden of multidrug resistance.
- *VanA* and *VanB* mechanisms: Replace the usual D-Alanine–D-Alanine (D-Ala-D-Ala) peptidoglycans, to which the vancomycin binds, with the abnormal D-Alanine–D-lactate (D-Ala-D-Lac)



Diagnosis

- **Gram stain**: Gram-positive cocci in pairs or chains
- **Catalase**: Usually negative
- **Culture:** on blood agar- mostly non-hemolytic
- **Identification:** of species ID by Vitek 2 system or validated system.
- **Antibiotic sensitivities**: assess vancomycin and high-level aminoglycoside resistance when relevant.



Treatment: Enterococcus urinary infection

- First decide: symptomatic cystitis, systemic UTI or colonization?
- Localized cystitis: choose an active urinary drug based on susceptibility.
- Susceptible Enterococcus may be treated with amoxicillin or nitrofurantoin in selected cases.
- Severe/systemic disease: systemic-active therapy guided by AST and specialist advice

Avoid treating asymptomatic bacteriuria outside guideline exceptions.

Case 1: Dysuria in a young woman

A 22-year-old woman develops burning urination, frequency and suprapubic pain. She is afebrile and has no flank pain. Urine culture yields Gram-positive cocci in clusters that are coagulase negative and novobiocin resistant.

- What is the most likely causative organism?
- What type of infection is this, and which drug is suitable?

Case 2: Catheter and Enterococcus in urine

A 70-year-old hospitalized patient has an indwelling urinary catheter. A urine culture grows *Enterococcus faecium*. The patient is afebrile and has no urinary symptoms or other evidence of infection.

- Does the culture prove symptomatic UTI?
- Should antibiotics be started immediately?
- What non-antibiotic intervention should be considered?

Take-home messages

- *S. saprophyticus* is an important community urinary pathogen despite being coagulase negative.
- Enterococci cause urinary and invasive infections, especially with healthcare or device exposure.
- Interpret cultures together with the patient’s symptoms and infection site.
- VRE colonization and asymptomatic bacteriuria usually do not need antibiotics.

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